

Contrast Before Answering: Train-Free Visual Pairing and Region Self-Verification for VLM Spatial Failures

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Abstract

Vision-language models can answer many image questions directly, yet visual binding, spatial relation, and count-like decisions may change when the relevant contrast is made explicit. We study a train-free inference scaffold, contrast-region self-verification (CRSV), that reorganizes inputs into paired contrasts or deterministic region views and asks the same hosted VLM backend to bind evidence before selecting an answer. The draft evaluation scope is MMVP and RefSpatial; OmniSpatial and PixMo-Count are outside the current result scope. Because the empirical evaluation is not yet complete, the headline result is **TBD** and every final comparison remains **TBD**. The draft therefore specifies the method, baselines, ablations, and evidence contract without claiming a completed outcome.

Introduction

The working question is whether a visual-language model can recover from under-specified single-image prompting when the same frozen model is shown a more diagnostic test-time view. The draft narrows this question to train-free visual contrast and spatial self-verification: no model weights, adapters, learned rewards, or training contracts are part of the method.

MMVP motivates the paired side of the question because its public benchmark construction pairs images around visual shortcomings such as attribute, orientation, presence, and counting-like distinctions (Tong et al. 2024). RefSpatial motivates the independent transfer side because it tests spatial reference and localization rather than paired multiple-choice contrast. The current benchmark scope is limited to these two families. OmniSpatial and PixMo-Count are not counted as result evidence in this draft.

This paper draft is intentionally fail-closed. The final MMVP pair-accuracy claim is **TBD**; the RefSpatial transfer claim is **TBD**; the matched-budget baseline comparison is **TBD**; and the ablation mechanism claim is **TBD**. These outcome statements remain unfilled until the evaluation is complete.

Related Work

Public VLM benchmarks have repeatedly exposed weaknesses in visual binding, compositional reasoning, and spatial interpretation. MMVP focuses on visual patterns that

can be easy for humans but brittle for multimodal models (Tong et al. 2024). Earlier spatial-reasoning probes also question whether language-vision systems reliably encode relations such as up, down, left, and right (Kamath, Hessel, and Chang 2023). Diagnostic synthetic and real-image benchmarks such as CLEVR, Visual Genome, and GQA provide longer-running context for compositional visual reasoning (Johnson et al. 2016; Krishna et al. 2016; Hudson and Manning 2019).

Recent work also explores adding structure at inference time or using visual structures to address binding failures (Miranda et al. 2025; Izadi et al. 2025). CRSV is positioned as a train-free, benchmark-scored scaffold in the same broad direction: it changes the evidence presentation and verification prompt, not the underlying model. Counting and spatial benchmark papers motivate the broader error family, but the current scope does not yet support any PixMo-Count or OmniSpatial result claim (Qharabagh, Ghofrani, and Fountoulakis 2024; Stogiannidis, McDonagh, and Tsaftaris 2025; Jia et al. 2025).

Method

CRSV has three paper-facing components. First, for paired sources such as MMVP, the contrast-pair prompt presents the relevant image pair together and requires independent answer binding for Image A and Image B. Second, for single-image spatial tasks such as RefSpatial, deterministic region views expose a grid-like reference frame derived from the same input image. Third, a self-verification pass asks the model to keep or revise the first answer while citing the evidence used for the final decision.

The method is train-free and inference-only. It uses the same hosted API VLM backend with temperature fixed at zero and does not create learned weights, optimizer state, or a training contract. The only method changes are prompt organization, image/view presentation, and verification instructions.

Experimental Setup

The evaluation covers MMVP and RefSpatial. MMVP uses held-out public rows, with conditions for direct prompting, longer direct prompting, self-consistency, question-type prompting, CRSV, pair shuffle, and pair order swap. RefS-

Table 1: Planned outcome slots. Numerical outcomes are TBD pending completed evaluation.

Family	Condition group	Result
MMVP	Direct and matched-budget baselines	TBD
MMVP	CRSV	TBD
MMVP	Pair shuffle / order swap	TBD
RefSpatial	Direct and matched-budget baselines	TBD
RefSpatial	CRSV and ablations	TBD

patial uses direct prompting, longer direct prompting, self-consistency, question-type prompting, CRSV, region-only, verify-only, and no-evidence-citation variants.

The evaluated model/backend is the hosted gpt-5-mini API VLM, evaluated with deterministic decoding. The evaluation is full scale for the two-family draft scope and uses only inference-time changes to prompts and views. Until the evaluation is complete, result values remain TBD.

Results

Table 1 is a placeholder matrix, not a result. The paper does not claim that CRSV improves, matches, or underperforms any baseline. The final MMVP headline comparison is **TBD**; the final RefSpatial transfer comparison is **TBD**; cost and parse-failure summaries are **TBD**; and confidence intervals or significance tests are **TBD**.

Analysis / Ablation

The planned mechanism checks are pair shuffle and pair order swap for MMVP, plus region-only, verify-only, and no-evidence-citation variants for RefSpatial. These are designed to distinguish a visual contrast mechanism from label balance, answer-position bias, prompt verbosity, and second-pass verification effects. The mechanism verdict is **TBD** until completed ablation results exist for the planned conditions.

Failure Cases

The planned failure taxonomy separates visual attribute binding, direction or relation inversion, counting aggregation, answer extraction, and verifier overcorrection. Representative held-out examples are **TBD** so that qualitative analysis is selected only after the evaluation is complete.

Conclusion

This scaffold defines a bounded, train-free paper around CRSV on the current MMVP and RefSpatial scope. At this point the method and evaluation contract are specified, but the final result is **TBD**. The final conclusion should report the supported improvement, narrowed mixed result, or measured negative result after the evaluation is complete.

References

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Reproducibility Checklist

- Method scope: train-free inference-only; no optimizer, learned weights, RL, or training contract.
- Benchmark scope: MMVP and RefSpatial; OmniSpatial deferred; PixMo-Count excluded unless separately audited.
- Backend: hosted gpt-5-mini API VLM with deterministic decoding.
- Metrics: MMVP pair/exact accuracy and RefSpatial official scorer output.
- Final metrics: TBD pending completed evaluation.